

1  
 $\mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined by:  $L(x) = L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 3x_2 \\ 2x_1 - x_2 \end{bmatrix}$   
 is linear mapping and

$\mathbb{R}^3 \rightarrow \mathbb{R}^2$  defined by  $L'\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 - 2x_3 \\ x_2^2 + 1 \end{bmatrix}$   
 is not linear mapping.

Addition:  $L(\vec{u} + \vec{v}) = L(\vec{u}) + L(\vec{v})$

Homogenous:  $L(C \cdot \vec{u}) = C \cdot L(\vec{u})$

Addition:

$$L(\vec{u} + \vec{v}) = L\left(\begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}\right) = \begin{bmatrix} (u_1 + v_1) + 3(u_2 + v_2) \\ 2(u_1 + v_1) - (u_2 + v_2) \end{bmatrix}$$

$$L(\vec{u}) = \begin{bmatrix} u_1 + 3u_2 \\ 2u_1 - u_2 \end{bmatrix} \quad L(\vec{v}) = \begin{bmatrix} v_1 + 3v_2 \\ 2v_1 - v_2 \end{bmatrix}$$

$$L(u) + L(v) = \begin{bmatrix} u_1 + 3u_2 + v_1 + 3v_2 \\ 2u_1 - u_2 + 2v_1 - v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 + 3(u_2 + v_2) \\ 2(u_1 + v_1) - (u_2 + v_2) \end{bmatrix}$$

$$L(C \cdot \vec{u}) = C \cdot \begin{bmatrix} u_1 + 3u_2 \\ 2u_1 - v_2 \end{bmatrix} = C \cdot L(u)$$

As homogeneity is justified, It is a linear mapping.



$$L'\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 - 2x_3 \\ x_2^2 + 1 \end{bmatrix}$$

$$\text{let } u = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \quad v = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$L'(u) = \begin{bmatrix} 1+2-2 \cdot 0 \\ 2^2+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$L'(v) = \begin{bmatrix} 2+1-2 \cdot 0 \\ 1^2+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$L'(u) + L'(v) = \begin{bmatrix} 3 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$$

$$u+v = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix} \quad L'(u+v) = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$

$$L'(u+v) = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$

$$L'(u) + L'(v) = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$$

$$L'(u+v) \neq L'(u) + L'(v)$$

Therefore it is not linear mapping.



2.

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

Given:

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 7 \\ -3 \end{bmatrix}$$

$$\text{Find } T\left(\begin{bmatrix} 7 \\ 2 \\ 2 \end{bmatrix}\right)$$

$$T\left(\begin{bmatrix} 7 \\ 2 \\ 2 \end{bmatrix}\right) = 7 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 7 \\ -3 \end{bmatrix} =$$

$$= \begin{bmatrix} 14 \\ 7 \end{bmatrix} + \begin{bmatrix} 10 \\ 0 \end{bmatrix} + \begin{bmatrix} 14 \\ -6 \end{bmatrix} = \begin{bmatrix} 24 + 14 \\ 7 - 6 \end{bmatrix} =$$

$$= \begin{bmatrix} 38 \\ 1 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 7 \\ 2 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 38 \\ 1 \end{bmatrix}$$

Therefore it is not linear



3. Given  $L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 4x_1 - 7x_2 \\ -3x_1 - 2x_2 \\ -4x_1 - 3x_2 \end{bmatrix}$

Find the standard matrix.

linear mapping  $\mathbb{R}^2 \rightarrow \mathbb{R}^3$ , so basis

are  $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$   $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

the co-domain of  $\mathbb{R}^3$  standard matrix is  $3 \times 2$  matrix.

image of  $e_1 = L(e_1) = L\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{bmatrix} 4 \cdot 1 - 7 \cdot 0 \\ -3 \cdot 1 - 2 \cdot 0 \\ -4 \cdot 1 - 3 \cdot 0 \end{bmatrix}$   
 $= \begin{bmatrix} 4 \\ -3 \\ -4 \end{bmatrix}$

image of  $e_2 = L(e_2) = L\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{bmatrix} 4 \cdot 0 - 7 \cdot 1 \\ -3 \cdot 0 - 2 \cdot 1 \\ -4 \cdot 0 - 3 \cdot 1 \end{bmatrix}$   
 $= \begin{bmatrix} -7 \\ -2 \\ -3 \end{bmatrix}$   $A = (L(e_1) \ L(e_2)) = \begin{bmatrix} 4 & -7 \\ -3 & -2 \\ -4 & -3 \end{bmatrix}$

the standard matrix of linear mapping is:

$$\begin{bmatrix} 4 & -7 \\ -3 & -2 \\ -4 & -3 \end{bmatrix}$$



4

$L: \mathbb{R}^4 \rightarrow \mathbb{R}^4$  defined by:

$$L \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -3x_1 - 3x_2 + x_3 + 3x_4 \\ x_1 - x_2 - 5x_3 - 5x_4 \\ -4x_1 - x_2 - 5x_3 + x_4 \\ -4x_1 - 2x_2 + 2x_3 - 3x_4 \end{pmatrix} \quad \begin{array}{l} \text{determine} \\ \text{injective} \\ \text{or not.} \end{array}$$

$$A = \begin{bmatrix} -3 & -3 & 1 & 3 \\ 1 & -1 & -5 & -5 \\ -4 & -1 & -5 & 1 \\ -4 & -2 & 2 & -3 \end{bmatrix} \quad \begin{array}{l} \text{If the RREF} \\ \text{for has a} \\ \text{pivot in each} \\ \text{column, rank} = 4 \end{array}$$

$\text{Ker}(L) = \{0\}$   $L$  is injective.

$$\begin{bmatrix} -3 & -3 & 1 & 3 \\ 1 & -1 & -5 & -5 \\ -4 & -1 & -5 & 1 \\ -4 & -2 & 2 & -3 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -1 & -5 & -5 \\ -3 & -3 & 1 & 3 \\ -4 & -1 & -5 & 1 \\ -4 & -2 & 2 & -3 \end{bmatrix} \quad \begin{array}{l} R_2 + 3R_1 \rightarrow R_2 \\ R_3 + 4R_1 \rightarrow R_3 \end{array}$$

$$\begin{bmatrix} 1 & -1 & -5 & -5 \\ 0 & -6 & -14 & -12 \\ 0 & -5 & -25 & -19 \\ -4 & -2 & 2 & -3 \end{bmatrix} \xrightarrow{4R_1 + R_4 \rightarrow R_4} \begin{bmatrix} 1 & -1 & -5 & -5 \\ 0 & -6 & -14 & -12 \\ 0 & -5 & -25 & -19 \\ 0 & -6 & -10 & -23 \end{bmatrix} \quad -\frac{1}{6} R_2 \rightarrow R_2$$

$$\begin{bmatrix} 1 & -1 & -5 & -5 \\ 0 & 1 & \frac{7}{3} & 2 \\ 0 & -5 & -25 & -19 \\ 0 & -6 & -18 & -23 \end{bmatrix} \quad \begin{array}{l} R_2 + R_1 \rightarrow R_1 \\ 5R_2 + R_3 \rightarrow R_3 \\ R_4 + 6R_2 \rightarrow R_4 \end{array} \quad \begin{bmatrix} 1 & 0 & -\frac{8}{3} & -3 \\ 0 & 1 & \frac{7}{3} & 2 \\ 0 & 0 & -\frac{10}{3} & -9 \\ 0 & 0 & -6 & -11 \end{bmatrix}$$



$$-\frac{3}{10} R_3 \rightarrow R_3 \quad \left[ \begin{array}{cccc} 1 & 0 & -\frac{8}{3} & -3 \\ 0 & 1 & \frac{7}{3} & 2 \\ 0 & 0 & 1 & \frac{27}{10} \\ 0 & 0 & -6 & -11 \end{array} \right] \quad \begin{array}{l} +R_1 \\ \frac{8}{3} R_3 \rightarrow R_1 \\ 6R_3 + R_4 \rightarrow R_2 \\ -\frac{7}{3} R_2 + R_2 \rightarrow R_2 \end{array}$$

$$\left[ \begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

RREF is Identity matrix.  
Each column has pivot.  
Matrix is invertible and  
transformation is  
injective.



5.

$$A = \begin{pmatrix} -7 & 1 & -4 \\ 1 & -7 & -4 \\ -4 & -4 & -4 \end{pmatrix}$$

Find characteristic polynomial of matrix

$$\det(A - \lambda I)$$

$$A - \lambda I = \begin{bmatrix} -7-\lambda & 1 & -4 \\ 1 & -7-\lambda & -4 \\ -4 & -4 & -4-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -7-\lambda & 1 & -4 \\ 1 & -7-\lambda & -4 \\ -4 & -4 & -4-\lambda \end{vmatrix}$$

$$-7-\lambda \begin{vmatrix} -4 & -4-\lambda \end{vmatrix} - 1 \begin{vmatrix} 1 & -4 \\ -4 & -4-\lambda \end{vmatrix} + (-4) \begin{vmatrix} 1 & -7-\lambda \\ -4 & -4 \end{vmatrix}$$

$$= (-7-\lambda)(-4-\lambda) - (-4)(-4)$$

$$= (7+\lambda)(4+\lambda) - 16$$

$$= \lambda^2 + 11\lambda + 28 - 16 = \lambda^2 + 11\lambda + 12$$

$$= (-7-\lambda)(\lambda^2 + 11\lambda + 12)$$

$$\begin{vmatrix} 1 & -4 \\ -4 & -4-\lambda \end{vmatrix} = 1 \cdot (-4-\lambda) - (-4)(-4) \\ = -4 - \lambda - 16 = -\lambda - 20 \\ (-1) \cdot (-\lambda - 20) = \lambda + 20$$



$$\begin{vmatrix} 1 & -7-\lambda \\ -4 & -4 \end{vmatrix} = 1 \cdot (-4) - (-4)(-7-\lambda) =$$

$$= -32 + 4\lambda = -32 - 4\lambda$$

$$-4(-32 - 4\lambda) = 128 + 16\lambda$$

$$-(\lambda+7)(\lambda^2+11\lambda+12) + (\lambda+20) + (128+16\lambda)$$

$$= (\lambda^3+11\lambda^2+11\lambda+7\lambda^2+77\lambda+84$$

$$= -\lambda^3-18\lambda^2-89\lambda-84+\lambda+20+128+16\lambda$$

$$= 17\lambda+148$$

$$= \lambda^3-18\lambda^2-72\lambda+64$$

$$P(\lambda) = \lambda^3 + 18\lambda^2 + 72\lambda - 64$$